

Matrix Algebra

Introduction

Matrix Notation

If A is an $m \times n$ matrix—that is, a matrix with m rows and n columns—then the scalar entry in the i th row and j th column of A is denoted by a_{ij} and is called the (i, j) -entry of A .

$$\begin{array}{c} \text{Column } j \\ \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ \text{Row } i & a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} = A \\ \begin{array}{ccccc} \uparrow & & \uparrow & & \uparrow \\ \mathbf{a}_1 & & \mathbf{a}_j & & \mathbf{a}_n \end{array} \end{array}$$

➤ a_{ij} is the i th entry (from the top) of the j th column vector $\mathbf{a}_j$.

➤ Each column of A is a list of m real numbers, which identifies a vector in $\mathbb{R}^m$.

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n]$$

➤ The **diagonal entries** in an $m \times n$ matrix $A = [a_{ij}]$ are $a_{11}, a_{22}, a_{33}, \dots$

➤ A **diagonal matrix** is a square $n \times n$ matrix whose nondiagonal entries are zero. An example is the $n \times n$ identity matrix, I_n .

Matrices

Dimensions of a Matrix

It is simply the number of rows and columns of a matrix, written in the form rows $\times$ columns.

Example:

The matrix below has 2 rows and 3 columns, so its dimensions are 2×3 . This is read aloud, "two by three."

$$A = \begin{bmatrix} 2 & 0 & -6 \\ 8 & 1 & 4 \end{bmatrix}$$

Matrices

Square Matrix

A matrix with equal numbers of rows and columns.

Example:

B is a 2×2 square matrix, whereas C is a 3×3 square matrix.

$$B = \begin{bmatrix} 4 & 3 \\ 6 & 2 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 6 & 4 \\ 9 & 3 & 0 \\ 7 & 8 & 1 \end{bmatrix}$$

Matrices

Equality of Matrices

Two matrices A & B are equal [written as $A = B$], if

1. they have the same size (i.e., the same number of rows and the same number of columns)
2. their corresponding entries are equal.

Matrices

Negative of a Matrix

- It is also called as the **Additive Inverse** of a Matrix.
- The matrix obtained by changing the sign of every matrix element.
- The additive inverse of matrix A is written $-A$.

$$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

$$-A = (-1)A = (-1) \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = \begin{bmatrix} -a & -b & -c \\ -d & -e & -f \end{bmatrix}$$

Matrices

Negative of a Matrix

Example:

$$-\begin{bmatrix} 2 & -4 \\ 7 & 10 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ -7 & -10 \end{bmatrix}$$

These are the calculations:

$-(2) = -2$	$-(-4) = +4$
$-(7) = -7$	$-(10) = -10$

Matrices

Matrix Operations

Matrices support various mathematical operations

- Addition of Matrices
- Subtraction of Matrices
- Scalar Multiplication of Matrices
- Multiplication of Matrices

Matrix Operations

Addition of Matrices

If A and B are $m \times n$ matrices, then sum $A + B$ is defined only when A and B are the same size, each entry in $A + B$ is the sum of the corresponding entries in A and B .

Consider the two matrices, A and B , of order 2×2 . Then, the sum is given by:

$$\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{bmatrix}$$

Matrix Operations

Addition of Matrices - Example

To add two matrices: add the numbers in the matching positions:

$$\begin{bmatrix} 3 & 8 \\ 4 & 6 \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 1 & -9 \end{bmatrix} = \begin{bmatrix} 7 & 8 \\ 5 & -3 \end{bmatrix}$$

These are the calculations:

$3+4=7$	$8+0=8$
$4+1=5$	$6-9=-3$

The two matrices must be the same size, i.e. the rows must match in size, and the columns must match in size.

Matrix Operations

Addition of Matrices

Note:

- The sum of a matrix and its additive inverse is the zero matrix.

OR

- If $A + B = 0 = B + A$, then B is called the additive inverse of A, and A is called the additive inverse of B.

Matrix Operations

Subtraction of Matrices

If A and B are two matrices of the same order, then we define

$$A - B = A + (-B)$$

Consider the two matrices, A and B, of order 2 x 2. Then, the difference is given by:

$$\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} - \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} = \begin{bmatrix} a_1 - a_2 & b_1 - b_2 \\ c_1 - c_2 & d_1 - d_2 \end{bmatrix}$$

Matrix Operations

Subtraction of Matrices - Example

To subtract two matrices: subtract the numbers in the matching positions:

$$\begin{bmatrix} 3 & 8 \\ 4 & 6 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 1 & -9 \end{bmatrix} = \begin{bmatrix} -1 & 8 \\ 3 & 15 \end{bmatrix}$$

These are the calculations:

$3-4=-1$	$8-0=8$
$4-1=3$	$6-(-9)=15$

*Note: subtracting is actually defined as the **addition** of a negative matrix: $A + (-B)$*

Matrix Operations

Scalar Multiplication

If A is a matrix and k any number, then the matrix which is obtained by multiplying the elements of A by k is called the scalar multiplication of A by k , and it is denoted by kA .

$$\triangleright \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\triangleright \quad kA = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

Matrix Operations

Scalar Multiplication - Example

We can multiply a matrix by a **constant** (*the value 2 in this case*):

The diagram illustrates the scalar multiplication of a constant (2) by a 2x2 matrix. A yellow circle containing the number 2 is multiplied by a matrix with elements 4, 0, 1, and -9. A yellow arrow points from the 2 to the 4 in the matrix, with the calculation $2 \times 4 = 8$ written above it. The resulting matrix has elements 8, 0, 2, and -18. The 8 and 2 in the resulting matrix are highlighted in yellow, corresponding to the 4 and 1 in the original matrix.

$$2 \times \begin{bmatrix} 4 & 0 \\ 1 & -9 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 2 & -18 \end{bmatrix}$$

These are the calculations:

$2 \times 4 = 8$	$2 \times 0 = 0$
$2 \times 1 = 2$	$2 \times -9 = -18$

We call the constant a **scalar**, so officially this is called "scalar multiplication".

Matrix Operations

EXAMPLE Let

$$A = \begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix}$$

Then

$$A + B = \begin{bmatrix} 5 & 1 & 6 \\ 2 & 8 & 9 \end{bmatrix}$$

but $A + C$ is not defined because A and C have different sizes. ■

$$2B = 2 \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{bmatrix}$$

$$A - 2B = \begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{bmatrix} = \begin{bmatrix} 2 & -2 & 3 \\ -7 & -7 & -12 \end{bmatrix} ■$$

Matrix Operations

EXAMPLE

Q. Find x and y ; given, $2x + 3y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix}$ and $3x + 2y = \begin{bmatrix} 2 & -2 \\ -1 & 5 \end{bmatrix}$.

Solution:

$$2x + 3y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} \dots (i)$$

$$3x + 2y = \begin{bmatrix} 2 & -2 \\ -1 & 5 \end{bmatrix} \dots (ii)$$

Multiplying (i) by 3 and (ii) by 2, we get

$$6x + 9y = \begin{bmatrix} 6 & 9 \\ 12 & 0 \end{bmatrix} \dots (iii)$$

$$6x + 4y = \begin{bmatrix} 4 & -4 \\ -2 & 10 \end{bmatrix} \dots (iv)$$

Matrix Operations

Solution: (Continued...)

Subtracting (iv) from (iii), we get

$$5y = \begin{bmatrix} 6 - 4 & 9 + 4 \\ 12 + 2 & 0 - 10 \end{bmatrix} = \begin{bmatrix} 2 & 13 \\ 14 & -10 \end{bmatrix} \Rightarrow y = \begin{bmatrix} \frac{2}{5} & \frac{13}{5} \\ \frac{14}{5} & \frac{-10}{5} \end{bmatrix} \Rightarrow y = \begin{bmatrix} \frac{2}{5} & \frac{13}{5} \\ \frac{14}{5} & -2 \end{bmatrix}$$

Putting the value of y in (iii), we get

$$2x + 3 \begin{bmatrix} \frac{2}{5} & \frac{13}{5} \\ \frac{14}{5} & -2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} \\ \Rightarrow 2x = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} - \begin{bmatrix} \frac{6}{5} & \frac{39}{5} \\ \frac{42}{5} & -6 \end{bmatrix} = \begin{bmatrix} 2 - \frac{6}{5} & 3 - \frac{39}{5} \\ 4 - \frac{42}{5} & 0 + 6 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} & -\frac{24}{5} \\ -\frac{22}{5} & 6 \end{bmatrix} \Rightarrow x = \begin{bmatrix} \frac{2}{5} & -\frac{12}{5} \\ -\frac{11}{5} & 3 \end{bmatrix}$$

Matrix Operations

Properties of Fundamental Matrix Operations

If r is a scalar and A is a matrix, then the **scalar multiple** rA is the matrix whose columns are r times the corresponding columns in A .

Let A , B , and C be matrices of the same size, and let r and s be scalars.

a. $A + B = B + A$

b. $(A + B) + C = A + (B + C)$

c. $A + 0 = A$

d. $r(A + B) = rA + rB$

e. $(r + s)A = rA + sA$

f. $r(sA) = (rs)A$

Matrix Operations

Dividing a Matrix

And what about division? Well we **don't** actually divide matrices, we do it this way:

$$A/B = A \times (1/B) = A \times B^{-1}$$

where **B⁻¹** means the "inverse" of B.

So we don't divide, instead we **multiply by an inverse**.